F1: Engine Logic & Technical Whitepaper

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Problem Definition

Input:

• A weighted directed graph $G = (V, E)$ with a non-negative weight function $w: E \to \mathbb{R}^+$, representing the regional map and road distances.
• A start node $s \in V$ and a destination node $t \in V$.
• A subset of nodes $S \subseteq V$ representing available EV charging stations.
• A maximum battery capacity (range constraint) $X \in \mathbb{R}^+$.

Output:

The absolutely lightest **Range-Feasible Path** P from s to t such that the total distance $w(P)$ is minimized, or a formal termination state indicating that the destination is unreachable under the current constraints.

Algorithm Description

Definition (Range-Feasible Path). A path P from s to t in G is called range-feasible (or safe) if it can be partitioned into a sequence of continuous subpaths $P = p_1 \cdot p_2 \cdot \ldots \cdot p_k$ such that the endpoints of each p_i belong to $\{s, t\} \cup S$, and the weight of each subpath satisfies $w(p_i) \leq X$.

Let $G' = (V', E')$ be a condensed virtual graph representing only the viable charging hubs, defined as:
• $V' = \{s, t\} \cup S$
• $E' = \{(u, v) \mid u, v \in V'$ and the lightest path between them in G has weight $\leq X\}$

1. Build G' by executing Dijkstra's Algorithm on G from every node in $\{s\} \cup S$. For every reachable node, if the lightest distance $d_u(v) \leq X$, add a directed edge (u, v) to E' with weight $w'(u, v) = d_u(v)$. (During this step, cache the physical sequence of nodes that generated this distance).

2. Execute Dijkstra's Algorithm starting from s strictly on the condensed graph G' to find the absolute lightest path P' to t.

3. If a path P' is found, reconstruct the expanded path P in the original graph G by replacing each virtual edge $e' \in P'$ with its cached physical subpath from Step 1.

4. Return P. If Step 2 yields no valid P', return a failure state indicating that the unit cannot safely arrive at t.

Proof of Correctness

Lemma 1: If no path P' is found in Step 2, then no range-feasible path exists in G.

Proof: Suppose for the sake of contradiction that a range-feasible path P from s to t does exist in G. By definition, P passes through a valid sequence of charging stations: $s \rightarrow S_1 \rightarrow S_2 \rightarrow \ldots \rightarrow S_k \rightarrow t$. Because P is range-feasible, the weight of every subpath between these consecutive nodes is strictly $\leq X$.

During Step 1, our algorithm evaluates the absolute lightest distances between all pairs in V'. Since the subpaths in P have weights $\leq X$, the mathematically lightest paths between these exact nodes must also have weights $\leq X$. Therefore, the corresponding edges (s, S_1), (S_1, S_2), ..., (S_k, t) were guaranteed to be generated and added to G'. This strictly implies that a valid path exists in G' from s to t, which contradicts our premise that no path P' was found. Thus, if the algorithm yields no P', no safe path exists. ■

Claim: Let P'' be the absolute lightest range-feasible path from s to t in G. The expanded path P returned by the algorithm is optimal, meaning $w(P) = w(P'')$.

Proof: Let the optimal path P'' traverse the specific sequence of station nodes $v_0, v_1, \ldots, v_k \in V'$ where $v_0 = s$ and $v_k = t$. Since P'' is safe, the weight of the subpath between any v_i and v_{i+1} is $\leq X$.

By the construction rules of G' in Step 1, there exists an edge $(v_i, v_{i+1}) \in E'$ with weight $w'(v_i, v_{i+1})$ equal to the true lightest path distance in G. Therefore, there exists a mapped path P'_{alt} in G' corresponding to the exact station sequence of P'', and its total weight in G' is $\leq w(P'')$.

Because Step 2 executes Dijkstra's algorithm on G', it is mathematically guaranteed to find the path P' with the global minimum weight in G'. Thus, $w'(P') \leq w'(P'_{\text{alt}}) \leq w(P'')$.

Since the expanded path P is formed by replacing the edges of P' with their physical routes in G, the weight maps perfectly: $w(P) = w'(P')$. Therefore, $w(P) \leq w(P'')$. Because P'' is defined fundamentally as the lightest possible safe path, it must be that $w(P) = w(P'')$. The algorithm guarantees optimal truth. ■

Time Complexity Analysis

Step 1 Graph Condensation:
Dijkstra's Algorithm is executed from s and from every node in S, a total of $|S| + 1$ independent runs. Each run on G with a minimum heap costs $O(|E| \log |V|)$. Step 1 therefore takes $O(|S| \cdot |E| \log |V|)$.

Step 2 Route Optimization:
A single run of Dijkstra on G' which has $|S| + 2$ nodes and at most $O(|S|^2)$ edges. This costs $O(|S|^2 + |S| \log |S|)$.

Step 3 Path Expansion:
Reconstructing the final route by walking cached subpaths is bounded by $O(|V|)$.

Total:
$\boldsymbol{O(|S| \cdot |E| \log |V| + |S|^2)}$

In a real-world regional road network the number of charging stations is far smaller than the total number of intersections ($|S| \ll |V|$).

Under these conditions the $O(|S|^2)$ term is negligible and the complexity is dominated entirely by the condensation phase, yielding an effective runtime of: $\boldsymbol{O(|S| \cdot |E| \log |V|)}$